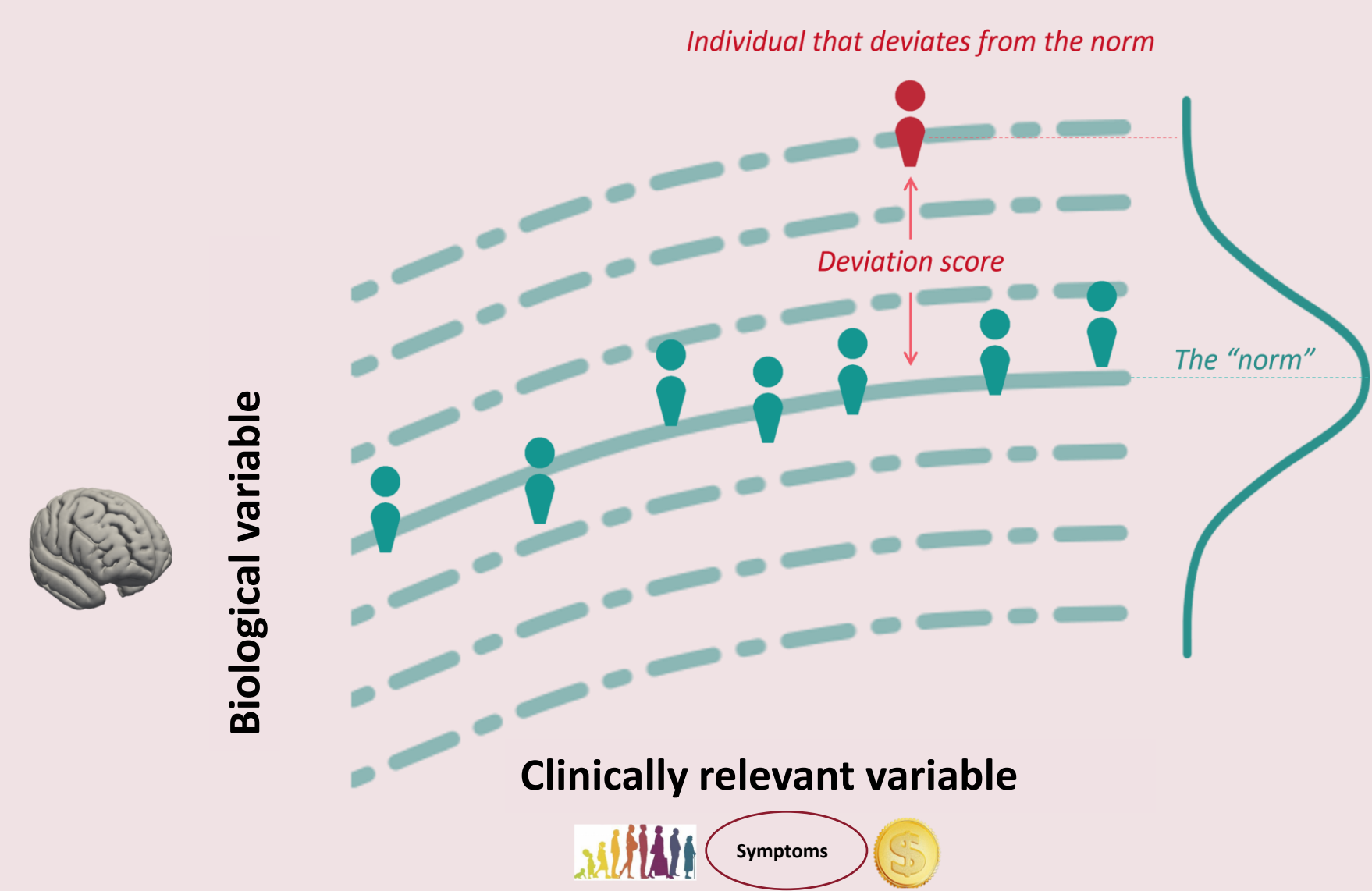


PCNtoolkit: open-source Python package for Normative Modelling

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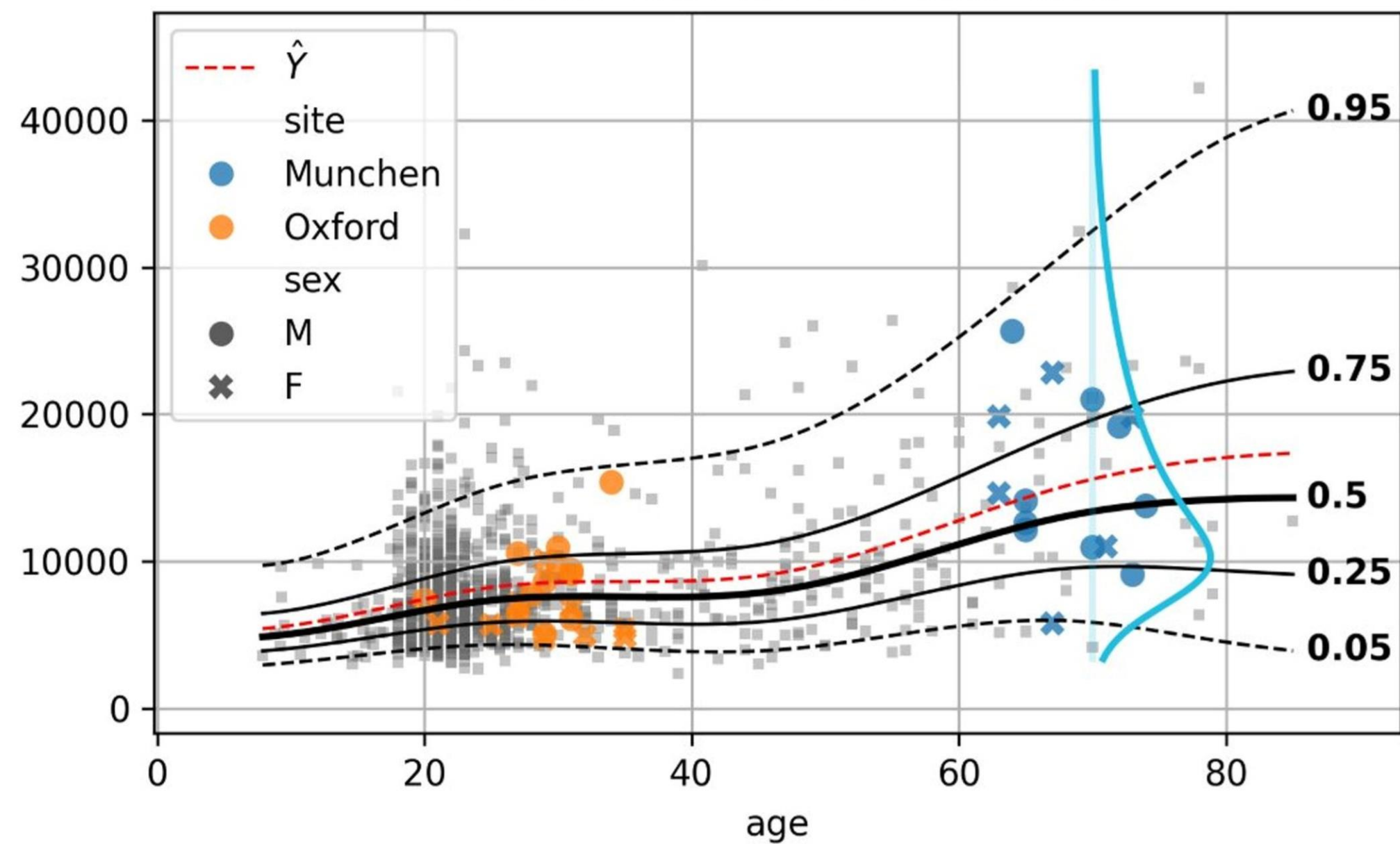
Why Normative Modelling?

- Psychiatric patients are **heterogeneous**: the same disorder can differ per person.
- Traditional methods compare patients to controls, asking how they differ **on average**.
- Normative modelling** instead embraces this heterogeneity and asks: "How does this individual deviate from a healthy reference population?"
- PCNtoolkit** makes this accessible in an open-source, reproducible and user-friendly way.

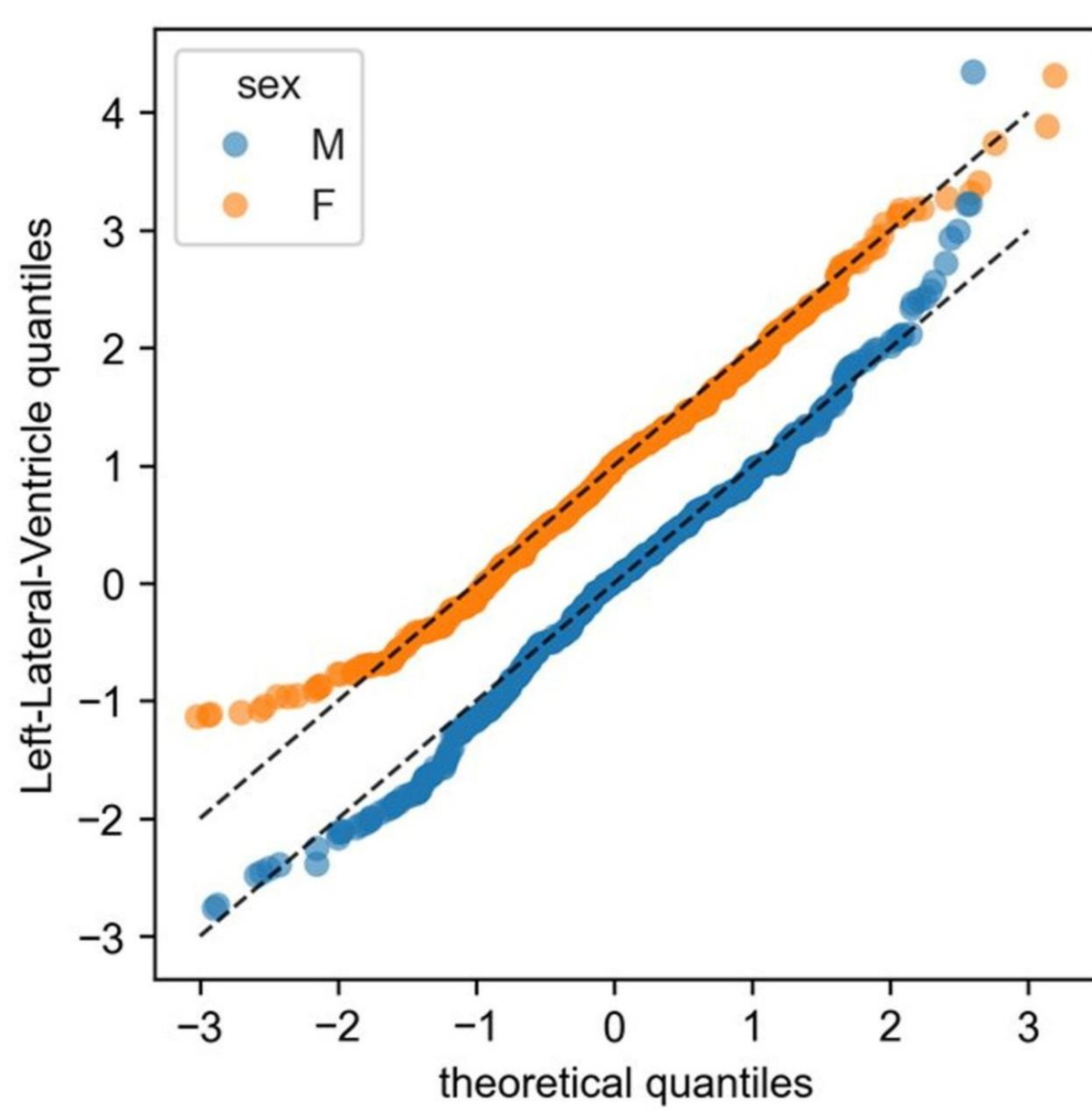
① Advanced models & plotting

Brain volume (mm³) – Left lateral ventricle

With harmonized fcon1000 data



Q-Q plot



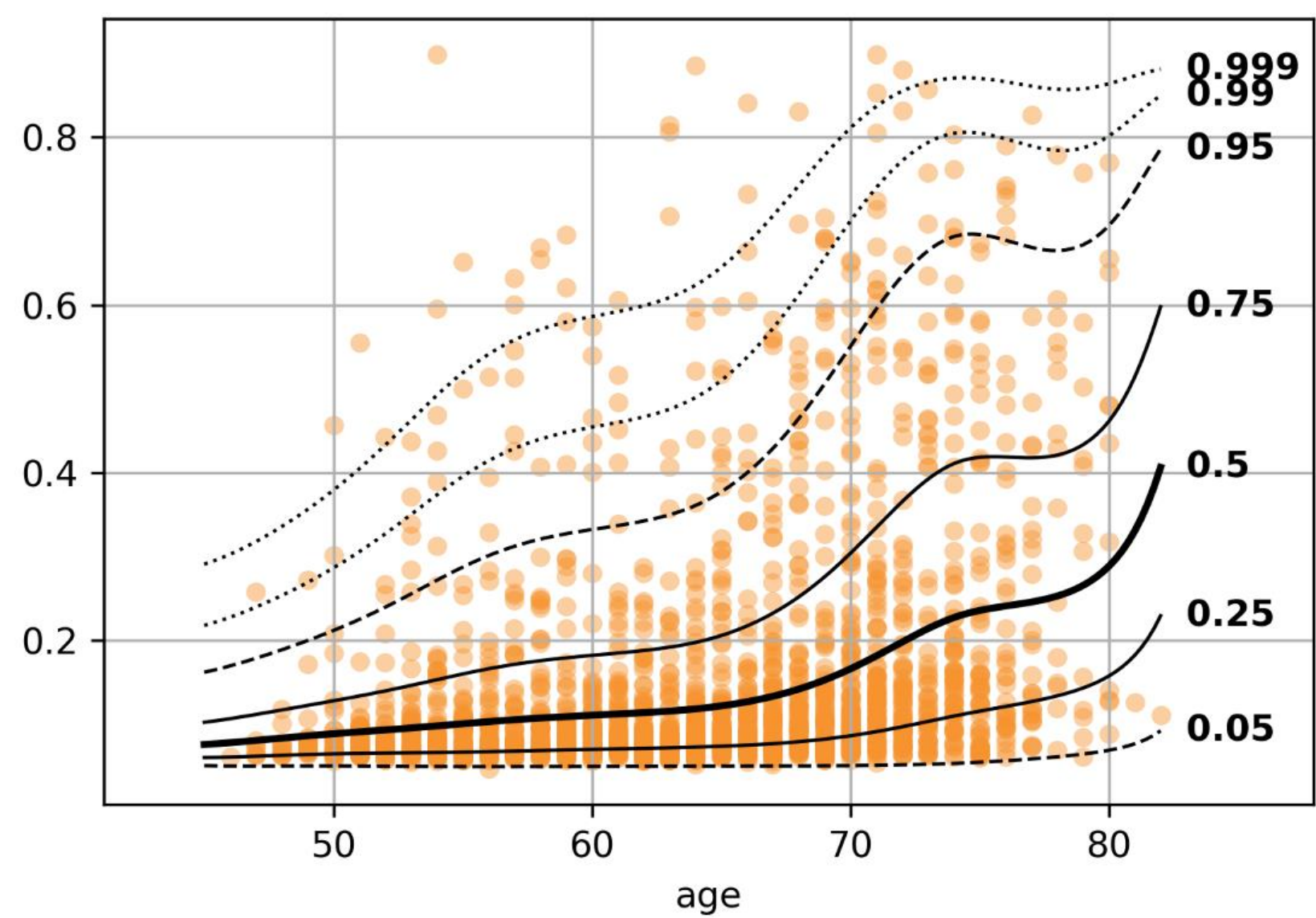
- Hierarchical Bayesian Regression with **SHASHb likelihood** [1,2] flexibly captures non-Gaussian distributions by modelling their skewness and kurtosis.

Flexible plotting with PCNtoolkit:

- Overlay** population centiles with site and sex data.
- Show the **conditional distribution** at any age.
- Built-in **Q-Q plots** for evaluation.

② Advanced models II

Fractional anisotropy – Fornix

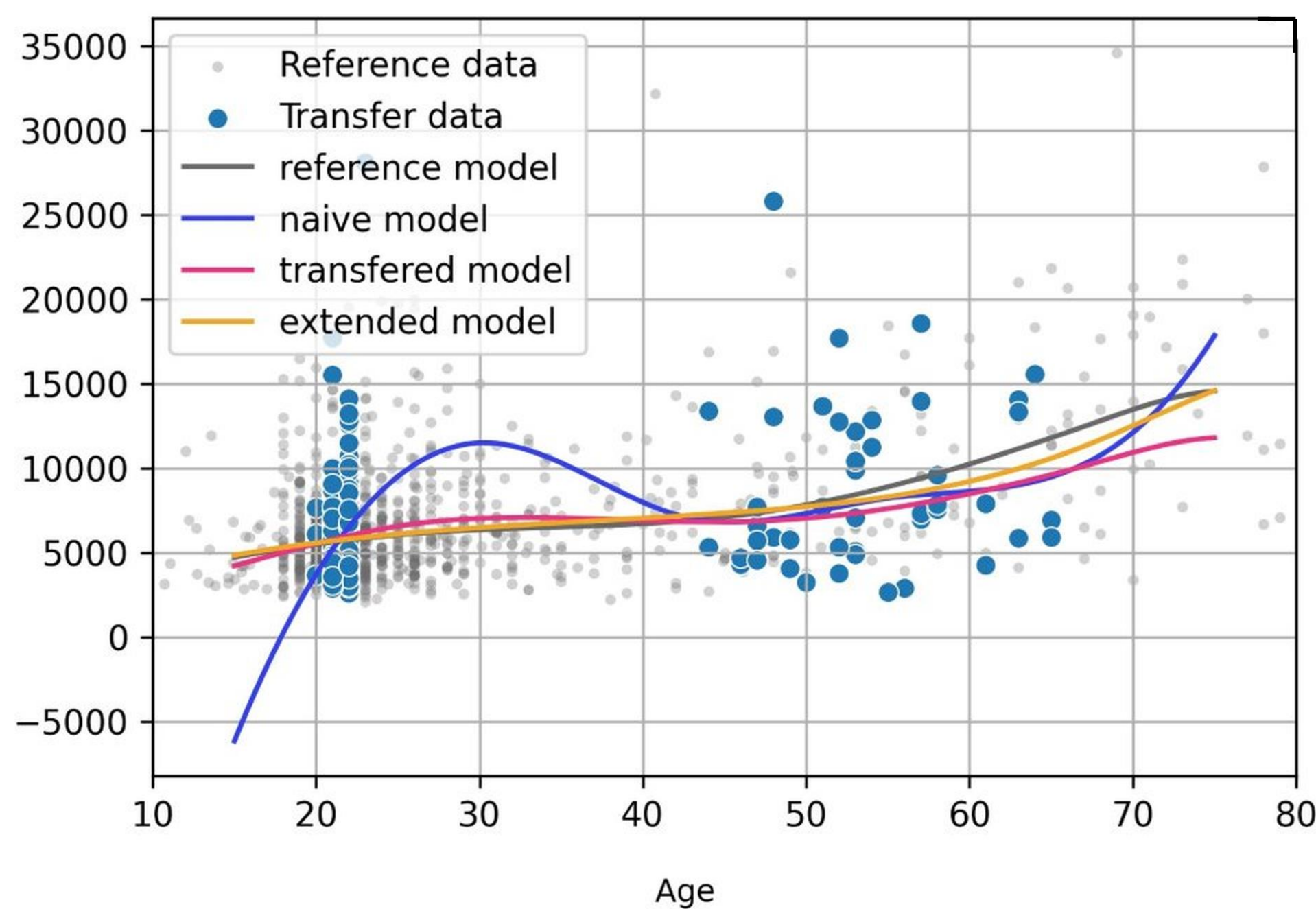


- Hierarchical Bayesian Regression with **Beta likelihood** [2] can model bounded (0–1) data.

[1] Biswal et al. (2010), *PNAS*
[2] de Boer et al. (2024), *Imaging Neuroscience*

③ Federated learning

Brain volume (mm³) - right lateral ventricle

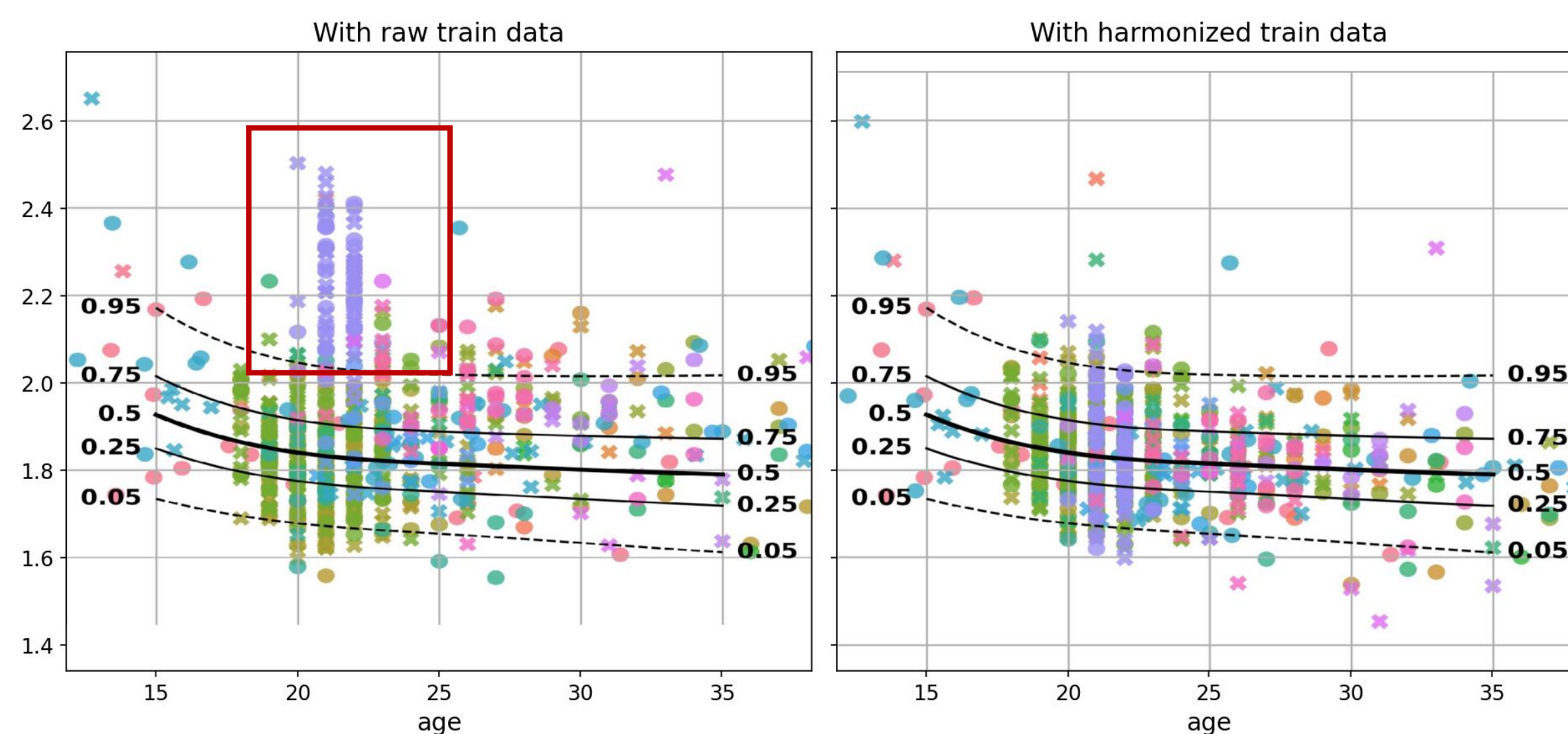


[3] Kia et al. (2022), *PLOS One*

- Federated learning** adapts a pre-trained reference model to new data *without access to the original training data*.
- Critical for **privacy-sensitive data** such as clinical neuroimaging data.
- Three types of federated learning: **transfer, extend, and merge** [3].
- A naïve model fit on a small dataset alone can be unreliable. By adapting a reference model to this small dataset, the resulting adapted (transferred & extended) models fit the data better.

④ Harmonization

Cortical thickness (mm) – Right cuneus gyrus

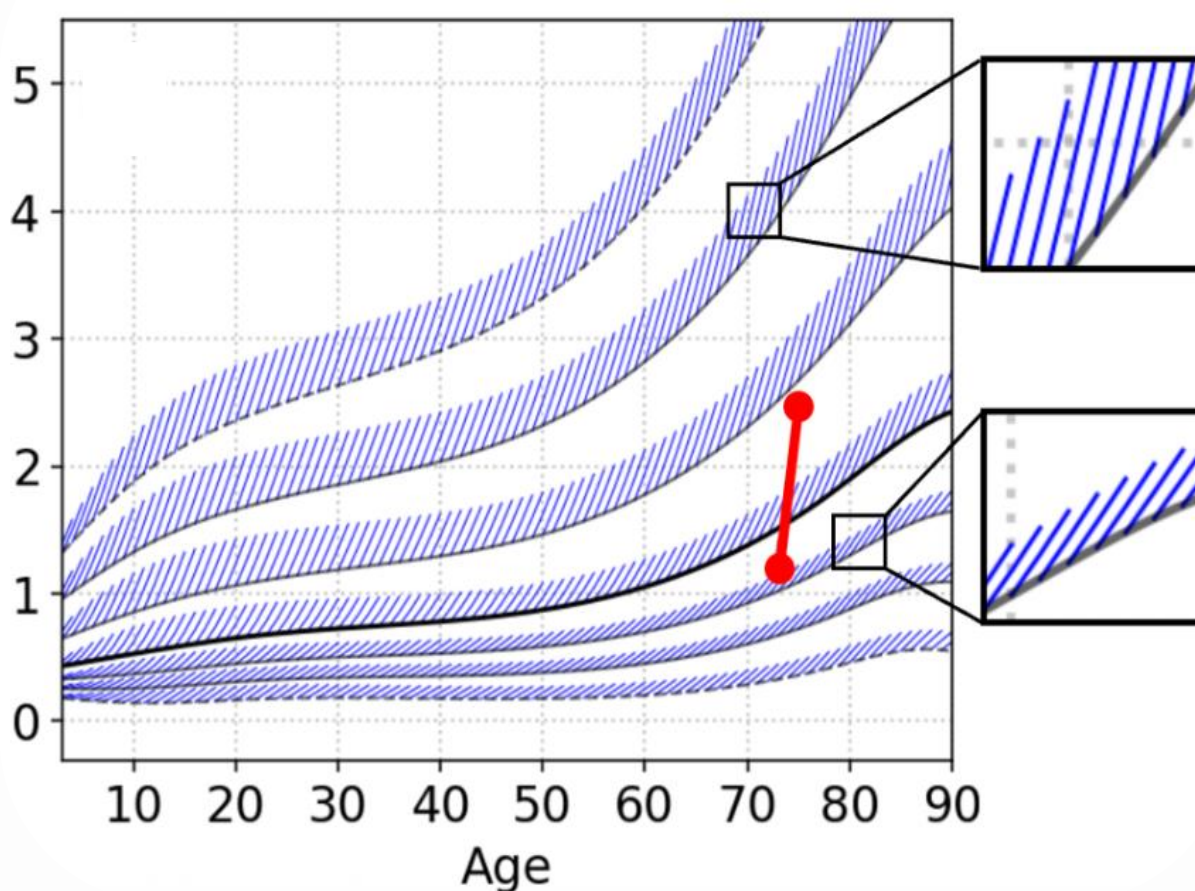


- PCNtoolkit can **harmonize** [4] data from many different groups (e.g., sites, sexes) onto a single common reference group.
- After harmonization, the data follow the centiles much more closely.

[4] Bayer et al. (2022), *Frontiers in Neurology*

⑤ Longitudinal modelling

Left-Lateral-Ventricle



[5] Bučková et al. (2025), *eLife*
[6] Bayer et al. (2026), *preprint*

- Normative modelling** evaluates an individual at *one time point*.
- However, two people can have different trajectories: one stable, the other diverging.
- Longitudinal modelling** [5,6] can reveal these diverging paths.
- It tracks individual changes **over time** relative to the expected trajectory.
- Thrive lines** [6] (blue lines) show this expected trajectory.
- Crossing centiles** can signal abnormal change (red line).
- z-diff** [5] and **z-gain** [6] scores statistically detect centile crossings.

⑥ PCNtoolkit ecosystem

OPEN-ACCESS PRE-TRAINED MODELS



Structural MRI



Diffusion MRI



Functional MRI

COMMUNITY RESOURCES



Website



Publications & Conferences



Neurostars forum



Google Summer of Code

CLINICAL APPLICATIONS

Schizophrenia

Depression

Alzheimer's